In microscope, the object distance is just more than the focal length f and the image is formed at distance v as shown in the Figure 7.38. Hence, f in equation (7.54) is replaced by v.

$$r_o = \frac{1.22\lambda v}{a} \quad (7.65)$$

If the distance between the two points on the object to be resolved is d_{min}, then the magnification m is,

$$m = \frac{r_o}{d_{\min}} \quad (7.66)$$

$$d_{\min} = \frac{r_o}{m} = \frac{1.22\lambda v}{am} = \frac{1.22\lambda v}{a(v/u)} = \frac{1.22\lambda u}{a} \quad [\because m = v/u]$$

$$d_{\min} = \frac{1.22\lambda f}{a} \quad [\because u \approx f] \quad (7.67)$$

On the object side,

$$2\tan\beta \approx 2\sin\beta = \frac{a}{f} \therefore \quad [a = f2\sin\beta] \quad (7.68)$$

$$d_{\min} = \frac{1.22\lambda}{2\sin\beta} \quad (7.69)$$

To further reduce the value of d_{min} the optical path of the light is increased by immersing the objective of the microscope into a bath containing oil of refractive index n.

$$d_{\min} = \frac{1.22\lambda}{2n\sin\beta} \quad (7.70)$$

Such an objective is called oil immersed objective. **The term $n\sin\beta$ is called *numerical aperture NA*.**

$$d_{\min} = \frac{1.22\lambda}{2(NA)} \quad (7.71)$$

The resolving power R_M of microscope is,

$$R_M = \frac{1}{d_{\min}} = \frac{2(NA)}{1.22\lambda} = \frac{2n\sin\beta}{1.22\lambda} \quad (7.72)$$

7.6.1.4. Resolving power of telescope

The resolving power of telescope is the reciprocal of the spatial resolution already derived in equation (7.54).

$$R_T = \frac{1}{r_0} = \frac{a}{1.22\lambda f} \quad (7.73)$$

7.6.2 Compound microscope

The diagram of a compound microscope is shown in Figure 7.39. The lens near the object is called as ***objective***. It forms a real, inverted and magnified image of the object. This serves as the object for the lens close to the eye called as ***eyepiece***. The eyepiece serves as a simple microscope that produces finally an enlarged and virtual image. The first inverted image formed by the objective is to be adjusted within the focus of the eyepiece so that the final image is formed nearly at infinity (or) at the near point. The final image is inverted with respect to the object.

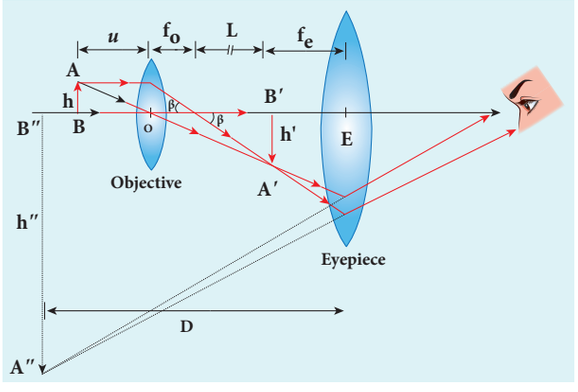


Figure 7.39 Compound microscope

7.6.2.1. Magnification in compound microscope

The lateral magnification produced by the objective is given by the equation (6.66),

$$m_o = \frac{h'}{h}$$

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